Performance Test

Date	10 October 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	F1 Race Intelligence Dashboard
Maximum Marks	

Project Performance Testing:

The project team shall fill the following information in the Project performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	5 tables loaded, 13,748 records in total: CIRCUITS 29, DRIVER_STANDINGS 311, LAP_TIMES 3,814, QUALIFYING 3,130, RACE_RESULTS 6,464 (42 columns). Size: add from Snowflake. (Page 2)
2.	Data Preprocessing & Quality	Null check: 396 blank finish_position / positions_gained / gap_to_leader in RACE_RESULTS (all DNF rows). Duplicate check: 334 duplicate race-driver rows (Carlos Sainz appears twice per race). (Page 2)
3.	New Fields Created	Any New Fields created
4.	Filter Efficiency Test	Filters used + comparison of query performance with different filters
5.	Dashboard & Visualization Testing	No of Visualizations / Graphs - **8 (+ 4 KPI cards)** 2 dashboards: Dashboard 2 (4 KPI cards + 4 charts), Dashboard 3 (4 charts). (Pages 3-4)
6	Query History	Query with Highest Execution time

Annexure A - Dataset Summary and Data Quality Checks

Table	Rows	Columns	Key columns
CIRCUITS	29	9	circuit_id, circuit_name, country, type, laps
DRIVER_STANDINGS	311	10	season, driver, team, points, wins, champion
LAP_TIMES	3,814	14	race_id, lap_number, lap_time_s, tyre_compound, drs_active
QUALIFYING	3,130	16	race_id, qualifying_position, q1/q2/q3 lap times
RACE_RESULTS	6,464	42	race_id, driver, team, grid/finish position, dnf, pit stops
TOTAL	**13,748**		

Data quality findings (RACE_RESULTS)

- Nulls: finish_position, positions_gained and gap_to_leader are blank in 396 rows; these are exactly the 396 rows with dnf = 1, and dnf_reason is filled for all of them (Engine 73, Hydraulics 70, Mechanical 67, Brake failure 64, Gearbox 62, Collision 60).
- No blanks found in grid_position, team or avg_pit_stop_time_s.
- Duplicates: 334 race-driver combinations appear twice. In every race, Carlos Sainz is listed under two teams, so each race has 20 rows (226 races) or 18 rows (108 races) but only 19 distinct drivers.
- Coverage: 334 distinct races, 19 distinct drivers, 10 teams. This agrees with the 6,464 race outcomes on the Tableau KPI card.
- QUALIFYING contains placeholder names (Driver_1 to Driver_10, Team_1 to Team_10) in many rows; these should be standardised or excluded.
- Counts are read from the exported dataset PDF; confirm them against SELECT COUNT(*) in Snowflake.

Annexure B - Dashboard Screenshots

Fig. 1 - Dashboard 2: F1 Race Intelligence Dashboard

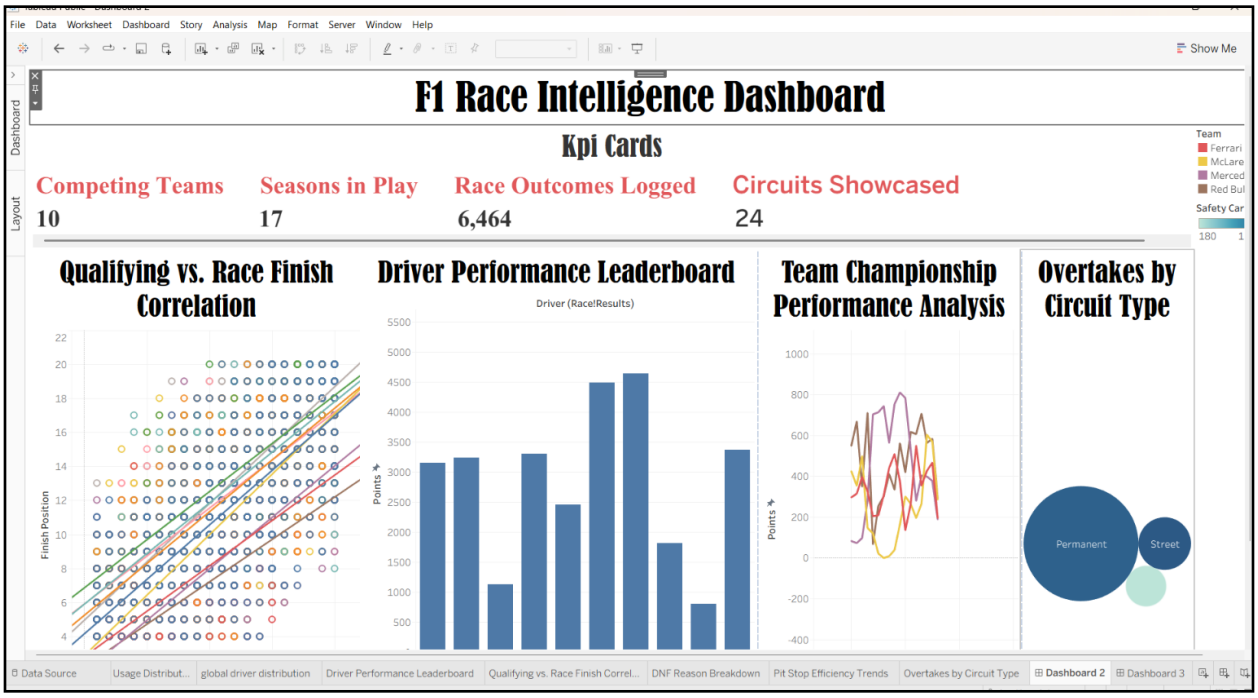


Fig. 2 - Dashboard 3: Pit Stop, Tyre Compound, DNF Reason and Global Driver Distribution

